PharmAchieve

OBESITY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

AE	Adverse Effect
BBS	Bardet-Biedl Syndrome
BMI	Body Mass Index
CI	Contraindicated
COC	Combined Oral Contraceptive
GIP	Glucose-Dependent Insulinotropic Polypeptide
GLP-1	Glucagon-like Peptide-1
HCP	Healthcare Professional
HDL	High-Density Lipoproteins
LDL	Low-Density Lipoproteins
LEPR	Leptin Receptor
MAOI	Monoamine Oxidase Inhibitor
MC4R	melanocortin 4 receptor
PCOS	Polycystic Ovary Syndrome
PCSK1	Proprotein Convertase Subtilisin/Kexin type 1
POMC	ProOpioMelanoCortin
SC	Subcutaneous
T2DM	Type 2 Diabetes Mellitus
WC	Waist Circumference
WHO	World Health Organization

OBJECTIVES

1. Identify the causes and risk factors for obesity
2. Understand how to calculate BMI
3. Assess the risks of obesity-associated conditions
4. Determine appropriate therapy for obesity

DEFINITION

Obesity is a global epidemic characterized by excess body fat

Tools to estimate and classify obesity:

- Adults: Body mass index (BMI) classification
- Children/teens: BMI-for-age percentile
- Waist circumference
- Edmonton Obesity Staging System

It is a chronic medical disease associated with an increased risk in mortality and several chronic conditions

BMI CLASSIFICATION FOR ADULTS

OBESITY CANADA GUIDELINES FOR BMI CLASSIFICATION

Caucasian, Europid, and North American Ethnicities

Classification		BMI (kg/m²)
Underweight		< 18.5
Normal/Healthy		18.5-24.9
Overweight		25.0-29.9
Obesity	Class 1	30.0-34.9
	Class 2	35.0-39.9
	Class 3	40.0-49.9
	Class 4	50.0-59.9
	Class 5	≥60.0

South-, Southeast-, or East Asian Ethnicities

Classification	BMI (kg/m²)
Underweight	<18.5
Normal/Healthy	18.5-22.9
Overweight – At Risk	23.0-24.9
Overweight – Moderate Risk	25.0-29.9
Overweight – Severe Risk	≥30.0

WAIST CIRCUMFERENCE

Waist circumference (WC) is an additional health risk assessment

- Patients have a higher risk of developing health conditions if:
  - Fat is mostly around waist instead of the hips
  - WC ≥ 102 cm (40 inches) in males
  - WC ≥ 88 cm (35 inches) in females

Limitations

- Only effective if BMI lies between 18.5 and 34.9
- Different ethnicities have different recommended WC cut-offs

How to measure waist circumference

- Wrap tape measure around the waist just above the hip bones
- Breathe out and measure

EDMONTON OBESITY STAGING SYSTEM

- Assess patients' medical, mental and functional symptoms
- Stage 0-4
  - Stage 0 corresponds to no limitations or abnormalities
  - Stage 1 corresponds to subclinical risks
  - Stage 2 corresponds to established disease
  - Stage 3 corresponds to severe disease
  - Stage 4 corresponds to end-stage disease

CAUSES/RISK FACTORS

Genetics

Drug Induced

- Antipsychotics
- Antidepressants
- Insulin, sulfonylureas
- Steroids
- COCs

Family History

- Presence of obesity in parents has been associated with obesity in the offspring

Health Conditions

- Hypothyroidism
- Cushing's Syndrome
- PCOS
- Pregnancy

Environmental Factors

- Social and cultural background
- Food prices
- Unavailability of healthy food

Emotional Factors

- Boredom
- Stress
- Depression

Dietary Habits

- High in fat and sugar

Eating Patterns

- Overeating/binging
- Frequency of eating

Sedentary lifestyle

Sleep deprivation

Smoking cessation

CLINICAL PRESENTATION

- Elevated BMI/waist circumference
- Requiring a larger size in clothes
- Changes in physical appearance
- Weight increase
- Comorbidities (Type 2 diabetes, hypertension, dyslipidemia, coronary artery disease, stroke, sleep apnea, certain cancers, osteoarthritis, gallbladder disease, difficulty breathing, depression, gout, poor self-image, non-alcoholic fatty liver disease)

DIAGNOSIS

By gathering a thorough history

- Physical examination
- Family history
- Social history
- Medication history

Laboratory investigations as needed

- Fasting blood glucose level
- Lipid profile
- Cholesterol (LDL cholesterol, HDL cholesterol, triglycerides)
- Blood pressure
- Screen for gout (e.g. uric acid level)
- Liver enzymes
- Amylase/Lipase
- Urinalysis

GOALS OF THERAPY

- ✓ Reduce body fat and prevent further weight gain
- ✓ Prevent health conditions associated with obesity
- ✓ Maintain a healthy weight
- ✓ Improve physical and psychosocial function

Reducing weight by 5–10% can result in important health benefits!

NON-PHARMACOLOGICAL MEASURES

- Aim for 30-60 minutes of moderate-vigorous physical activity on most days
- Try to include strength training a few times a week
- Maintain a food diary to assess healthy eating habits (ensure balanced meals and snacks throughout the day, instead of fewer, larger meals)
- Consider seeking advice from a dietician to help tailor meal plans
- Minimize sedentary activity such as watching television, scrolling on phone, playing video games, etc
- Bariatric surgery may be considered in some cases (BMI≥40 kg/m^2 or BMI ≥ 35 kg/m^2 with at least 1 adiposity-related disease); however, it can cause malabsorption of nutrients and the patient will require lifelong medical care
- Liposuction is a cosmetic procedure to reduce fat

PHARMACOLOGICAL THERAPY

Prescription Therapy

- Consider in patients:
  - BMI ≥ 30 kg/m^2 or ≥ 27 kg/m^2 with adiposity-related complications
  - Want to maintain weight loss achieved by behavioral changes and to prevent regain
  - Unsatisfied with their progress after lifestyle changes
- Pharmacotherapy must be used as an adjunct to lifestyle changes

PHARMACOLOGICAL THERAPY

Lipase Inhibitors

Drugs	Orlistat (Xenical®)
Dosing	120 mg once daily to TID
AEs	Oily discharge, flatus with discharge, fecal urgency, steatorrhea
Comments	Accompany with a meal containing fat Can decrease absorption of fat-soluble vitamins Contraindicated with chronic malabsorption syndrome or cholestasis

Appetite Suppressant/Opioid Antagonist

Drugs	Bupropion/naltrexone (Contrave®)
Dosing	Initial: 1 tablet daily x 7 days then 1 tablet BID x 7 days, then 2 tablets QAM and 1 tablet QHS x 7 days Maintenance: 2 tablets BID Moderate-severe renal impairment: 1 tablet BID Mild-moderate liver impairment: 1 tablet daily
AEs	Nausea, constipation, vomiting, headache, dizziness, insomnia, dry mouth
Comments	Contraindicated with concurrent opioid or MAOI use Each tablet contains 8 mg naltrexone and 90 mg bupropion

PHARMACOLOGICAL THERAPY

GLP-1 Receptor Agonists

Drugs	Semaglutide (Wegovy®) Semaglutide is also sold under the brand name Ozempic® • It is officially indicated for diabetes, despite being used for obesity in practice
Dosing	Titrate to 2.4 mg SC once weekly
AEs	Nausea, vomiting, constipation, diarrhea, rarely acute pancreatitis
Comments	CI: pregnancy, breastfeeding, personal or family history of medullary thyroid carcinoma, or multiple endocrine neoplasia syndrome (type 2)

Drugs	Liraglutide (Saxenda®)
Dosing	Titrate to 3 mg SC once daily
AEs	Nausea, vomiting, constipation, diarrhea, rarely acute pancreatitis
Comments	CI: pregnancy, breastfeeding, personal or family history of medullary thyroid carcinoma, or multiple endocrine neoplasia syndrome (type 2)

PHARMACOLOGICAL THERAPY

GIP/GLP-1 Receptor Agonist

Drugs	Tirzepatide (Zepbound®)
Dosing	Titrate to up to 15 mg SC once weekly
AEs	Nausea, vomiting, constipation, diarrhea, dyspepsia, rarely acute pancreatitis
Comments	CI: pregnancy, breastfeeding, personal or family history of medullary thyroid carcinoma, or multiple endocrine neoplasia syndrome (type 2) Can decrease the absorption of oral contraceptives

Melanocortin 4 Receptor (MC4R) agonist

Drug	Setmelanotide (Imcivree®) • Approved for treatment of obesity caused by specific rare genetic causes (i.e. BBS, or deficiency in POMC, PCSK1 or LEPR variants interpreted as non-benign)
Dosing	Titrate to up to 3 mg SC once daily
AEs	Nausea, vomiting, skin hyperpigmentation. Rare: sexual, mental health problems
Comments	CI: pregnancy, breastfeeding. Avoid use in personal or family history of melanoma or pre-melanoma skin lesions

5 A'S OF OBESITY MANAGEMENT – COUNSELLING POINTS

Ask	• Permission to discuss weight • Determine readiness to change
Assess	• Use BMI and WC to classify and evaluate risk • Assess potential causes
Advise	• Health risks of obesity • Appropriate treatment and therapy
Agree	• Realistic goals and weight management plan
Assist	• Address any barriers • Provide helpful resources • Regular follow-up

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	**Degree of Change**	**Frequency of monitoring**
Weight	≥ 5% weight loss after first 3 months ≥ 10% weight loss overall	Patient: Weekly HCP: monthly for the first 3 months, then quarterly thereafter
BMI	Normalize to < 30 kg/m^2 or < 27 kg/m^2 if weight-related complications	HCP: monthly for the first 3 months, then quarterly thereafter
Waist circumference	Normalize to < 88 cm in women and < 102 cm in men	
A1C	Normalize to < 7% *	Every 3 months
Fasting Blood glucose	4-7 mmol/L *	variable
Blood pressure	Normalize to < 130/80 mmHg *	Routinely

* Target may vary based on comorbidities

MONITORING PARAMETERS

SAFETY ENDPOINTS			
Parameter	**Degree of Change**	**Frequency of monitoring**	**Timeframe**
Adverse effects (e.g. nausea/vomiting, constipation, diarrhea)	Prevent or minimize	HCP: every month for first 3 months of therapy, then every 3 months	Duration of therapy
Acute pancreatitis	Prevent	Symptom-based screening HCP: every month for first 3 months of therapy, then every 3 months	

TAKEAWAY

- Manage expectations and set realistic goals
- Focus on a healthier lifestyle, not cosmetic appearance
- Dietary and lifestyle modifications are first-line therapy
- Medications indicated for chronic obesity management in Canada, in combination with non-pharmacological interventions, include orlistat, liraglutide, semaglutide, naltrexone/bupropion, tirzepatide and setmelanotide
- Avoid herbals, “fad diets” and other non-supported treatment options
- Be cautious of inappropriate medication use
  - Laxatives: induce bowel movements
  - Diuretics: increase production of urine

CASE STUDY

JA is a 32-year-old East African male who comes to your pharmacy asking for RestoraLAX® (Polyethylene Glycol 3350) as he can not find it on the shelves. You notice he is a bit anxious, so you invite him to sit down in the counselling room. J.A. tells you that he is a nursing student who is currently completing his nursing practice placement in the cardiology department at Michael Garron Hospital. He has been struggling with weight gain for the past several years, which has affected his self-confidence. His busy work and placement schedule has put him under a lot of stress, which has contributed to his weight problem. Last week, he had a follow-up appointment with his family doctor and found out his BMI was 42.3 kg/m^2, an increase from 41.2 kg/m^2 just 3 months ago. He tries to follow his doctor's recommendations and stay active as much as possible; he goes for weekly jogs whenever his schedule permits. His medical history is significant for depression and anxiety, for which he takes bupropion XL 150 mg once a day, and lorazepam 0.5 to 1 mg one to 2 times a day, as needed for anxiety. J.A. used to smoke 1 pack of cigarettes a day, but was able to successfully quit 2 months ago. He tells you his father also suffers from obesity and got recently diagnosed with type 2 diabetes, which he thinks is related, at least partially, to his underlying obesity issue. J.A.'s mother is a thyroid cancer survivor. J.A. is inquiring about the use of RestoraLAX® (Polyethylene Glycol 3350) as it was recommended by one of his work colleagues.

1. How many risk factors are most likely contributing to J.A.'s current weight gain issue?
   a. 2
   b. 3
   c. 4
   d. 5

CASE STUDY (CONTINUED)

JA is a 32-year-old East African male who comes to your pharmacy asking for RestoraLAX® (Polyethylene Glycol 3350) as he can not find it on the shelves. You notice he is a bit anxious, so you invite him to sit down in the counselling room. J.A. tells you that he is a nursing student who is currently completing his nursing practice placement in the cardiology department at Michael Garron Hospital. He has been struggling with weight gain for the past several years, which has affected his self-confidence. His busy work and placement schedule has put him under a lot of stress, which has contributed to his weight problem. Last week, he had a follow-up appointment with his family doctor and found out his BMI was 42.3 kg/m^2, an increase from 41.2 kg/m^2 just 3 months ago. He tries to follow his doctor's recommendations and stay active as much as possible; he goes for weekly jogs whenever his schedule permits. His medical history is significant for depression and anxiety, for which he takes bupropion XL 150 mg once a day, and lorazepam 0.5 to 1 mg one to 2 times a day, as needed for anxiety. J.A. used to smoke 1 pack of cigarettes a day, but was able to successfully quit 2 months ago. He tells you his father also suffers from obesity and got recently diagnosed with type 2 diabetes, which he thinks is related, at least partially, to his underlying obesity issue. J.A.'s mother is a thyroid cancer survivor. J.A. is inquiring about the use of RestoraLAX® (Polyethylene Glycol 3350) as it was recommended by one of his work colleagues.

2. What is your recommendation regarding the use of RestoraLAX® (Polyethylene Glycol 3350)?

a. It can be used but should not be the sole treatment option for obesity
b. It should not be used due to lack of supporting evidence
c. It can be used for weight loss only if the patient has comorbid constipation condition
d. It may be used initially as a temporary solution for weight loss until the patient gets a prescription for an appropriate medication

CASE STUDY (CONTINUED)

JA is a 32-year-old East African male who comes to your pharmacy asking for RestoraLAX® (Polyethylene Glycol 3350) as he can't find it on the shelves. You notice he is a bit anxious, so you invite him to sit down in the counselling room. J.A. tells you that he is a nursing student who is currently completing his nursing practice placement in the cardiology department at Michael Garron Hospital. He has been struggling with weight gain for the past several years, which has affected his self-confidence. His busy work and placement schedule has put him under a lot of stress, which has contributed to his weight problem. Last week, he had a follow-up appointment with his family doctor and found out his BMI was 42.3 kg/m², an increase from 41.2 kg/m² just 3 months ago. He tries to follow his doctor's recommendations and stay active as much as possible; he goes for weekly jogs whenever his schedule permits. His medical history is significant for depression and anxiety, for which he takes bupropion XL 150 mg once a day, and lorazepam 0.5 to 1 mg one to 2 times a day, as needed for anxiety. J.A. used to smoke 1 pack of cigarettes a day, but was able to successfully quit 2 months ago. He tells you his father also suffers from obesity and got recently diagnosed with type 2 diabetes, which he thinks is related, at least partially, to his underlying obesity issue. J.A.'s mother is a thyroid cancer survivor. J.A. is inquiring about the use of RestoraLAX® (Polyethylene Glycol 3350) as it was recommended by one of his work colleagues.

3. Upon inquiring further, JA shares that he is following a low-fat diet that he does not wish to change, and he also does not prefer to add more medications to his existing regimen.

   What is the most appropriate treatment option for J.A.?

   a. Orlistat (Xenical®)
   b. Bupropion/naltrexone (Contrave®)
   c. Semaglutide (Ozempic®)
   d. Liraglutide (Saxenda®)

CASE STUDY (CONTINUED)

J.A. is a 32-year-old East African male who comes to your pharmacy asking for RestoraLAX® (Polyethylene Glycol 3350) as he can't find it on the shelves. You notice he is a bit anxious, so you invite him to sit down in the counselling room. J.A. tells you that he is a nursing student who is currently completing his nursing practice placement in the cardiology department at Michael Garron Hospital. He has been struggling with weight gain for the past several years, which has affected his self-confidence. His busy work and placement schedule has put him under a lot of stress, which has contributed to his weight problem. Last week, he had a follow-up appointment with his family doctor and found out his BMI was 42.3 kg/m^2, an increase from 41.2 kg/m^2 just 3 months ago. He tries to follow his doctor's recommendations and stay active as much as possible; he goes for weekly jogs whenever his schedule permits. His medical history is significant for depression and anxiety, for which he takes bupropion XL 150 mg once a day, and lorazepam 0.5 to 1 mg one to 2 times a day, as needed for anxiety. J.A. used to smoke 1 pack of cigarettes a day, but was able to successfully quit 2 months ago. He tells you his father also suffers from obesity and got recently diagnosed with type 2 diabetes, which he thinks is related, at least partially, to his underlying obesity issue. J.A.'s mother is a thyroid cancer survivor. J.A. is inquiring about the use of RestoraLAX® (Polyethylene Glycol 3350) as it was recommended by one of his work colleagues.

4. What other interventions/recommendations are likely to help with further weight loss?
    a. Diet control and food portioning, while maintaining regular weekly jogs
    b. Diet control and food portioning, aerobic physical exercises and resistance training for at least 150 min/week
    c. Diet control and food portioning (e.g. ketogenic diet; refer to a dietitian for assistance)
    d. Given patient's current BMI, bariatric surgery should be considered as an option

REFERENCES

1. Rueda-Clausen CF, Poddar M, Lear SA, Poirier P, Sharma AM. Canadian Adult Obesity Clinical Practice Guidelines: Assessment of People Living with Obesity. Available from: https://obesitycanada.ca/guidelines/assessment. Updated August 4, 2020. Accessed September 2024.
2. Promoting Optimal Monitoring of Child Growth in Canada: Using the New WHO Growth Charts. Collaborative Statement from: Dietitians of Canada, Canadian Pediatric Society, The College of Family Physicians of Canada, Community Health Nurses of Canada. 2020. https://cps.ca/uploads/tools/growth-charts-statement-FULL.pdf.
3. Explore Overweight and Obesity. National Heart, Lung, and Blood Institute- NIH. Updated July 2012. Accessed August 2014.
4. 5A's of Obesity Management for Adults. Canadian Obesity Network. Accessed August 2014. http://www.obesitynetwork.ca/5As_adult.
5. Wharton S. Obesity. E-Therapeutics - Therapeutic Choices. Canadian Pharmacists Association. Updated June 12, 2024. Accessed September 2024.
6. Heschuk S. Nutrition: Weight Management. E-Therapeutics- Minor Ailments. Canadian Pharmacist Association. Updated March 31, 2020. Accessed December 2020.
7. Bray, G. Obesity in adults: Overview of management. In: UpToDate. Updated November 2016. Accessed February 2016.
8. Bray G. Obesity in adults: Prevalence, screening, and evaluation. In: UpToDate. Updated January 2017. Accessed February 2017.
9. Wharton S, Lau DCW, Vallis M et al. Obesity in adults: a clinical practice guideline. CMAJ. 2020 Aug 4;192(31):E875-E891.
10. Pedersen SD, Manjoo P, Wharton S. Pharmacotherapy for obesity management. Canadian Adult Obesity Clinical Practice Guidelines. October 2022. obesitycanada.ca
11. Health Canada. Dietary reference intakes tables: Reference values for macronutrients. Updated December 18, 2023. Accessed September 2024. https://www.canada.ca/en/health-canada/services/food-nutrition/healthy-eating/dietary-reference-intakes/tables/reference-values-macronutrients.html
12. Pedersen SD, Manjoo P, Dash S, Jain A, Pearce N, Poddar M. Pharmacotherapy for obesity management in adults: 2025 clinical practice guideline update. *CMAJ*. 2025;197(27):E797-E809. Published 2025 Aug 10. doi:10.1503/cmaj.250502
13. American Diabetes Association Professional Practice Committee; 8. Obesity and Weight Management for the Prevention and Treatment of Type 2 Diabetes: Standards of Care in Diabetes–2025. *Diabetes Care* 1 January 2025; 48 (Supplement_1): S167–S180. https://doi.org/10.2337/dc25-S008

CHANGE LOG

March 2025
- Formatting changes throughout

May 2025
- Slide 19: Correct answer changed from B to C (4 risk factors)

July 2025
- Content reviewed; no changes made

August 2025
- Slide 3: Legend updated
- Slide 17: New slide with newly approved drugs for obesity as per Canadian guidelines published on Aug 11, 2025
- Slide 19: Updated Takeaway
- References updated

October 2025
- Slide 19-20: Added monitoring parameters slides

January 2026
- Slide 2: Updated NAPRA competencies
- Slide 24: Updated question 3